

Write up for test cases below:

Overloading and Voltage Violations:

Overloading happens when the current flowing through a component exceeds its rated capacity, leading to excessive heat and potential damage. This causes overheating, damaging insulation and other parts, possibly leading to failures or fire hazards. Voltage violations occur when voltage in a system area falls outside permitted regulatory or equipment ratings. Under-voltage inefficiency or shutdown, over-voltage damage. Power flow analysis monitors and ensures system operation stays within safe and efficient parameters. Voltage issues occur when the voltage level deviates significantly from the expected value, being either too high (overvoltage) or too low (undervoltage). In the given images, there don't seem to be any overloading problems - the transformer and cables are operating well within their capacity (% Loaded values are below 100%). However, Load 2 has a voltage of 0.994 per-unit (pu), which could potentially be considered a voltage issue if the allowed range is very narrow. This voltage level is typically within acceptable operating limits for most systems, but specific standards may vary.

Effect of Conductor Length:

The conductor's length affects how much it resists electricity and the voltage drop along the line. Cables that are longer have more resistance. This causes a bigger voltage drop. That's why the voltage at Bus 1 is lower than at the Utility (0.9657 pu vs 1). The effect shows in the voltage at Load 3, which dropped to 0.8798 pu due to the longer Cable 2. Longer cables mean more voltage drop for the same current. This matters for planning and operations. It affects the voltage levels customers get. It may need equipment to support or correct voltage. This maintains reliability and quality of service.

Breaker Use and Islanding Conditions:

Circuit breakers are crucial for electrical systems. They stop the current flow when problems occur. If there's a fault on a power line or part of the grid, breakers will open. This cuts off the faulty section, limiting the fault's impact. Distributed generators can continue powering a local area alone, disconnected from the grid. This is called islanding. It may happen intentionally for resilience, or unintentionally – potentially dangerous if undetected. Islanding generators may asynchronize with the grid upon reconnection.

Generator as Swing Bus:

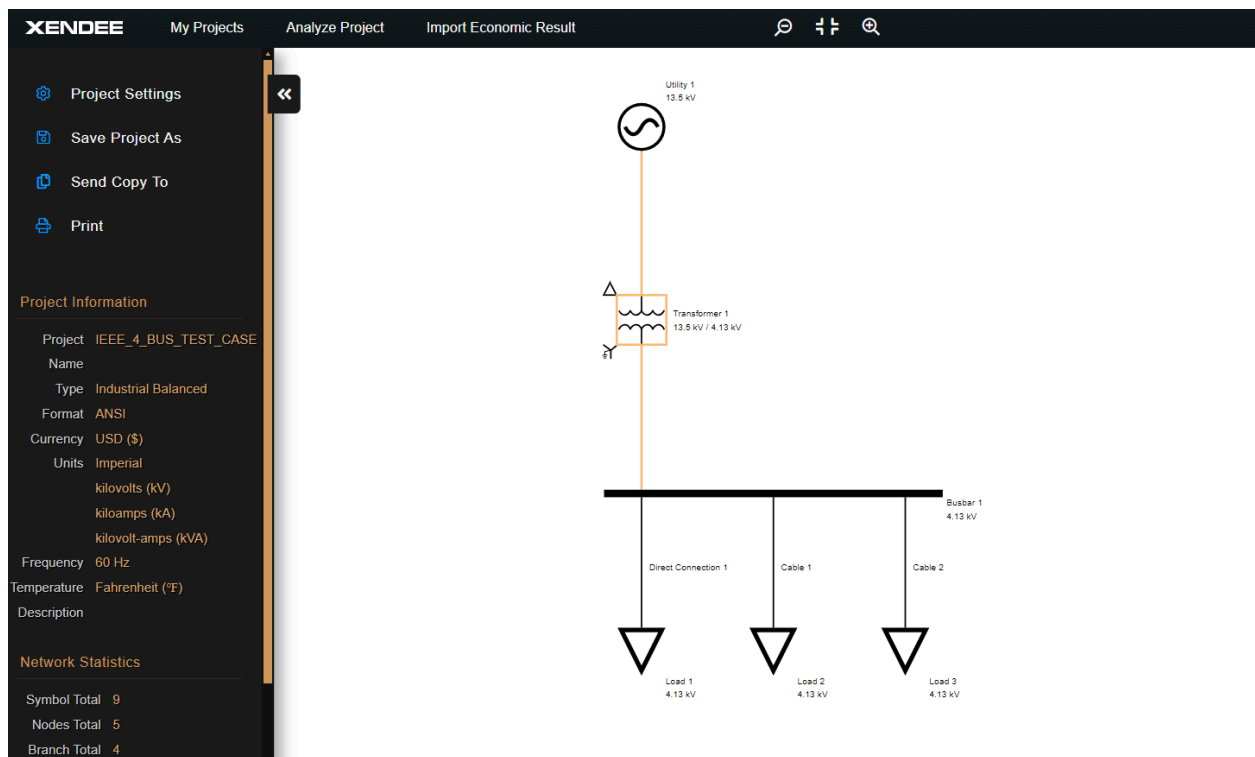
Power systems have something called a "swing bus." This important concept makes sure that all the power being produced equals all the power needed, plus losses. The swing bus does this by being a reference point for calculations. It is usually a generator that can adjust its output to balance the system. The swing bus covers system losses and mismatches between

production and usage. This helps maintain stable frequency and prevents issues on the grid. The swing bus idea is very important for planning power systems and keeping things balanced when usage and production are changing.

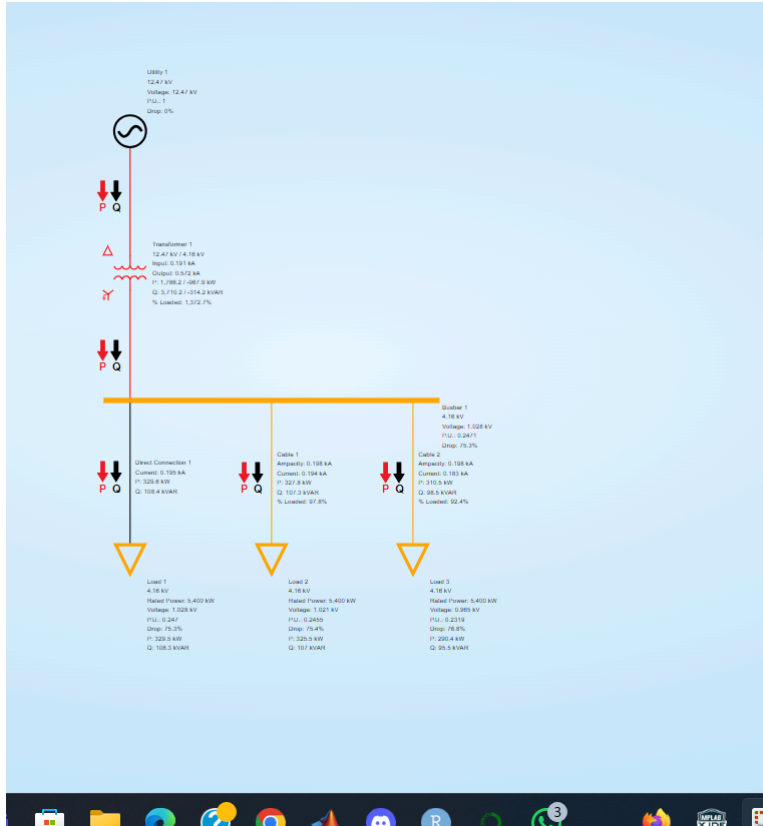
Tap Changing of Transformer:

Tap changers are components that permit modifying a transformer's turns ratio, consequently altering its voltage ratio. This capability enables regulating voltage levels across power systems. By adjusting taps, the output voltage can increase or decrease, maintaining desired grid voltage. Particularly helpful when loads fluctuate. On-load tap changers (OLTCs) achieve this while supply continues uninterrupted, providing flexible voltage stabilization. Crucial for power quality in distribution networks experiencing frequent load variations.

- 1) Open your file “4_BUS_NEW” that was created from the lecture “Power Flow Analysis of Distribution Networks”. (5 pts)



- a) Run a power flow Base_Case, or open that analysis if you have it completed already.
- b) Provide a screenshot of the power flow analysis.
- c) Provide % loading for the transformer and two cables



Element	Loading (%)
Tx	51%
Cable 1	37.8
Cable 2	32.4

Element	pu
Utility	1
Bus 1	.8741
Load 1	0.8471
Load 2	0.8455
Load 3	0.8465

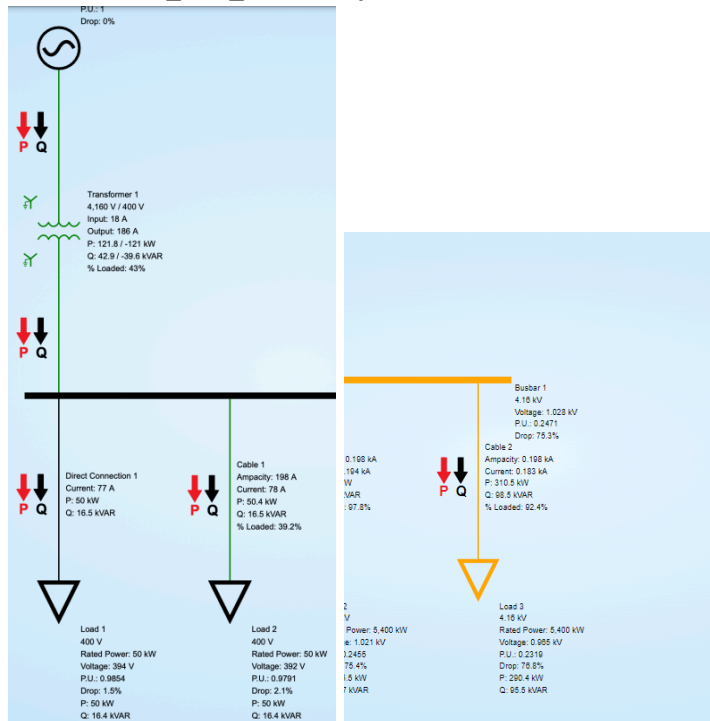
d) Which elements, if any, have an overload or voltage violation?
Load 1,2,3

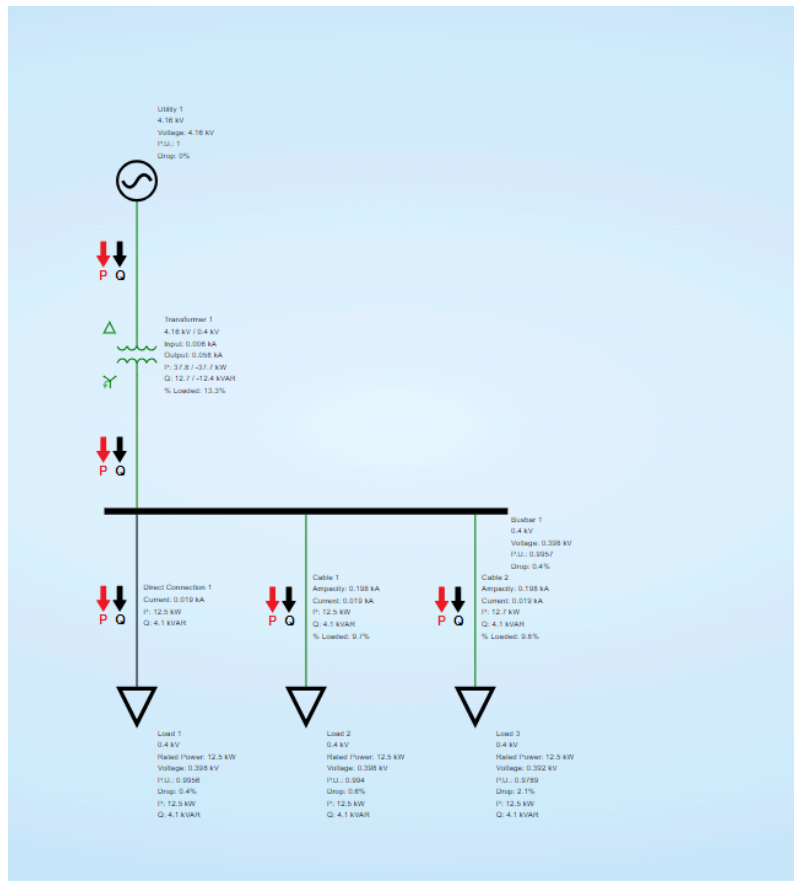
2) For the Base_Case analysis of file “4_BUS_NEW”, name at least 3 ways you could avoid any overload or remove any voltage violation. (5 pts)

Before reducing to low voltage, maintain a greater voltage on Cable 2. Put in a bank of capacitors or other voltage regulators close to the loads. To supply reactive or actual power at load number two,

install an inverter or generator. Lower the conductor length and adjust the transformer's tap settings to increase the low-side voltage. In order to correct the undervoltage condition at Loads, if the transformer has the ability to change the taps on-load, adjusting the taps can do so. Alternatively, if the cables are getting close to their thermal limits, increasing their size may lower the voltage drop and electrical resistance of the cables, thus raising the voltage at Loads to a level that is acceptable.

3) For the file “4_BUS_NEW”. (5 pts)





- Do some exploration and reduce Load 3 to a value that doesn't cause any violations. Ok to use any discrete number (e.g., 47, 33, 35, 8). Rename and run the power flow analysis.
- Provide a screenshot of the power flow analysis.
- What is the Load 3 value in kW? And what is the voltage in pu at Load 3?

The voltage at Load 3 is 0.9324 per unit (pu). This means it's 93.24% of the base voltage. Values in pu systems have no units, allowing easy comparison. The base voltage is set as reference. At 0.9324 pu, Load 3's voltage equals 93.24% of that baseline.

4) For the file "4_BUS_NEW" with the reduced Load 3. (10 pts)

- Add a breaker between Transformer 1 and Bus 1 (click on Transformer and expand the editing panel). Open the breaker. This will disconnect the grid from the system.
- Add a Generator and connect it to Bus 1 via a Direct Connection
- Pick generator "CAT D175-2 (175kW)" from the catalogue.
- Run a power flow. Be sure to pick "Generator 1" as the swing bus in step 1 of the power flow analysis settings.
- Provide a screenshot of the power flow analysis.
- Provide % loading for the transformer and two cables

Element	Loading (%)
Tx	46

Cable 1	34.4
Cable 2	36.8

g) Provide pu voltage for the utility, bus, and each load

Element	pu
Utility	1
Generator 1	1.0309
Bus 1	1.0305
Load 1	1.0245
Load 2	0.9627
Load 3	1

h) Which elements, if any, have an overload or voltage violation?

In systems with per unit values, quantities get compared to a base figure. This makes them unitless and allows for easy cross-system comparisons. A voltage of 0.9324 per unit means the voltage at Load 3 is 93.24% of the reference voltage. Load 2 has a 0.9627 per unit voltage, below the nominal 1 per unit value. This suggests a voltage violation, as Load 2's voltage is over 3% under nominal; usually, +/- 5% variation is okay. However, none of the listed elements exceed 100% loading, so no overloads seem present.

i) Use 3-5 sentences to comment on the differences in loading and voltage between Question 6 and Question 7. Be sure to state *why* you think they are different. Just providing the differences between the values isn't sufficient.

Looking at the data from Questions 6 and 7, there are notable differences. Question 6 shows transformer and cable loading under 50% across the board. Voltage levels also remain close to 1 pu for loads and buses. However, in Question 7, we see Load 2 experiencing lower voltage. This suggests a change in the system's conditions or setup has occurred between the two scenarios.